AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 29 · WORKBOOK

Performance and cost experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete matched-workload performance record.

Keep case IDs, durations, outcomes, timing scope, cost units, objective and remaining evidence gaps together.

Keep one habit: Recall → W1–W3/W5 → independent C/D case → performance project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Accessibility and user feedback. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Incident investigation. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Deployment environments. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Performance experiments compare **matched workload**s with declared timing boundaries and quality constraints. Averages can hide slow cases.

- Use the same cases and environment; record every duration and success. For a simple **median** sort values and average the middle pair for even counts. Keep failed requests visible.

Accessibility and user feedback

> **My remembered idea, example and correction**

Incident investigation

> **My remembered idea, example and correction**

Deployment environments

> **My remembered idea, example and correction**
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Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Performance experiments need matched workloads, timing boundaries, quality constraints and complete outcomes. Main supplied Q1–Q4 durations A [1, 1, 1, 9] seconds and B [2, 2, 2, 2], all pass. A total 12, mean 3, sorted middle pair 1/1, median 1, observed max 9; B total 8, mean/median/max 2. B lowers mean but A lowers median; no universal winner without an objective.

  Added duration scope is request start to final outcome, including failure records. Separate picture six cases each with supplied equivalent quality: A 18 seconds/12 cost units, B 12 seconds/18 units; means A 3 seconds/2 units, B 2 seconds/3 units. Changed budget is mean cost at most 2 units: A meets equality, B fails. Totals alone establish no individual cap, median or maximum. Units are fictional, not currency.

  Preserve failed outcomes, slow cases, case IDs, settings and timing scope. Added code keeps all durations in a separate 3/4-quality probe. Constructed arithmetic is not an actual benchmark, repeated-run study, stable tail estimate or current provider-price comparison.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Performance experiments need matched workloads, timing boundaries, quality constraints and complete outcomes. Main supplied Q1–Q4 durations A [1, 1, 1, 9] seconds and B [2, 2, 2, 2], all pass. A total 12, mean 3, sorted middle pair 1/1, median 1, observed max 9; B total 8, mean/median/max 2. B lowers mean but A lowers median; no universal winner without an objective.

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  Preserve failed outcomes, slow cases, case IDs, settings and timing scope. Added code keeps all durations in a separate 3/4-quality probe. Constructed arithmetic is not an actual benchmark, repeated-run study, stable tail estimate or current provider-price comparison.

- Durations A=[1, 1, 1, 9] seconds, B=[2, 2, 2, 2]. All pass. Compute mean and **median**. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Performance experiments need matched workloads, timing boundaries, quality constraints and complete outcomes. Main supplied Q1–Q4 durations A [1, 1, 1, 9] seconds and B [2, 2, 2, 2], all pass. A total 12, mean 3, sorted middle pair 1/1, median 1, observed max 9; B total 8, mean/median/max 2. B lowers mean but A lowers median; no universal winner without an objective.

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  Preserve failed outcomes, slow cases, case IDs, settings and timing scope. Added code keeps all durations in a separate 3/4-quality probe. Constructed arithmetic is not an actual benchmark, repeated-run study, stable tail estimate or current provider-price comparison.

- A report omits this qualification: Resource improvement under one workload is not universal efficiency or equal quality. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Performance experiments need matched workloads, timing boundaries, quality constraints and complete outcomes. Main supplied Q1–Q4 durations A [1, 1, 1, 9] seconds and B [2, 2, 2, 2], all pass. A total 12, mean 3, sorted middle pair 1/1, median 1, observed max 9; B total 8, mean/median/max 2. B lowers mean but A lowers median; no universal winner without an objective.

  Added duration scope is request start to final outcome, including failure records. Separate picture six cases each with supplied equivalent quality: A 18 seconds/12 cost units, B 12 seconds/18 units; means A 3 seconds/2 units, B 2 seconds/3 units. Changed budget is mean cost at most 2 units: A meets equality, B fails. Totals alone establish no individual cap, median or maximum. Units are fictional, not currency.

  Preserve failed outcomes, slow cases, case IDs, settings and timing scope. Added code keeps all durations in a separate 3/4-quality probe. Constructed arithmetic is not an actual benchmark, repeated-run study, stable tail estimate or current provider-price comparison.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- C=[1, 2, 3, 10], D=[3, 3, 3, 3]. Compute means/**median**s; one D response is wrong.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled performance and cost experiments **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of performance and cost experiments.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Performance and cost experiments. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Performance and cost experiments. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- C=[1, 2, 3, 10], D=[3, 3, 3, 3]. Compute means/**median**s; one D response is wrong.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Measure time and cost. Two fixed configurations answer the same six questions and meet the same quality rubric.
- A totals 18 seconds and 12 cost units. B totals 12 seconds and 18 units.
- Calculate per-case seconds and cost units for A and B. Which meets a budget of at most 2 cost units per case on average, and what speed trade-off does that choice make?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

AI ACADEMY / AI-6/29

Mean resource use per case

	Seconds/case	Cost units/case
A	3	2
B	2	3

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-6/29

Performance and cost experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute mean time

A averages 3 seconds; B averages 2 seconds per case.



2. Compute mean cost

A averages 2 units; B averages 3 units per case.



3. Compare trade-off

B saves one second but costs one extra unit per case.

CHANGE ONE THING

Require at most 2 cost units per case.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project milestone

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Measure time and cost. Two fixed configurations answer the same six questions and meet the same quality rubric.
- Which labelled step matches this detail? B saves one second but costs one extra unit per case.
- 1. Compute mean cost
- 2. Compute mean time
- 3. Compare trade-off
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compare trade-off B saves one second but costs one extra unit per case. A fair performance experiment holds the workload and measurement procedure fixed.

Faster is not automatically preferable under a budget.

Changed-case question and optional hint

- Changed-case practice: Require at most 2 cost units per case on average. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. A meets the budget; B does not.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: A meets the budget; B does not. The mandatory resource condition changes the decision.

Repair and retain the record

Review Accessibility and user feedback (ai-6/28). Goal: Explain the **mechanism** and terminology of accessibility and user feedback.

Short practice: Explain what each text, keyboard and assistive-technology change adds to the feedback. Why does changing the red circle to orange alone not solve the communication barrier?.

Return to this week and explain what you changed.

Review Incident investigation (ai-6/26). Goal: Explain the **mechanism** and terminology of incident investigation. Short practice: Order the three incident records and locate the old **evidence** version.

How does the investigation change if the trace instead retrieves version 6 but drafts room C?. Return to this week and explain what you changed.

Review Deployment environments (ai-6/23). Goal: Explain the **mechanism** and terminology of deployment environments. Short practice: Identify the staging data and credential boundary.

Why must the rehearsal stop if its credential reaches production even though the **process** is labelled staging?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

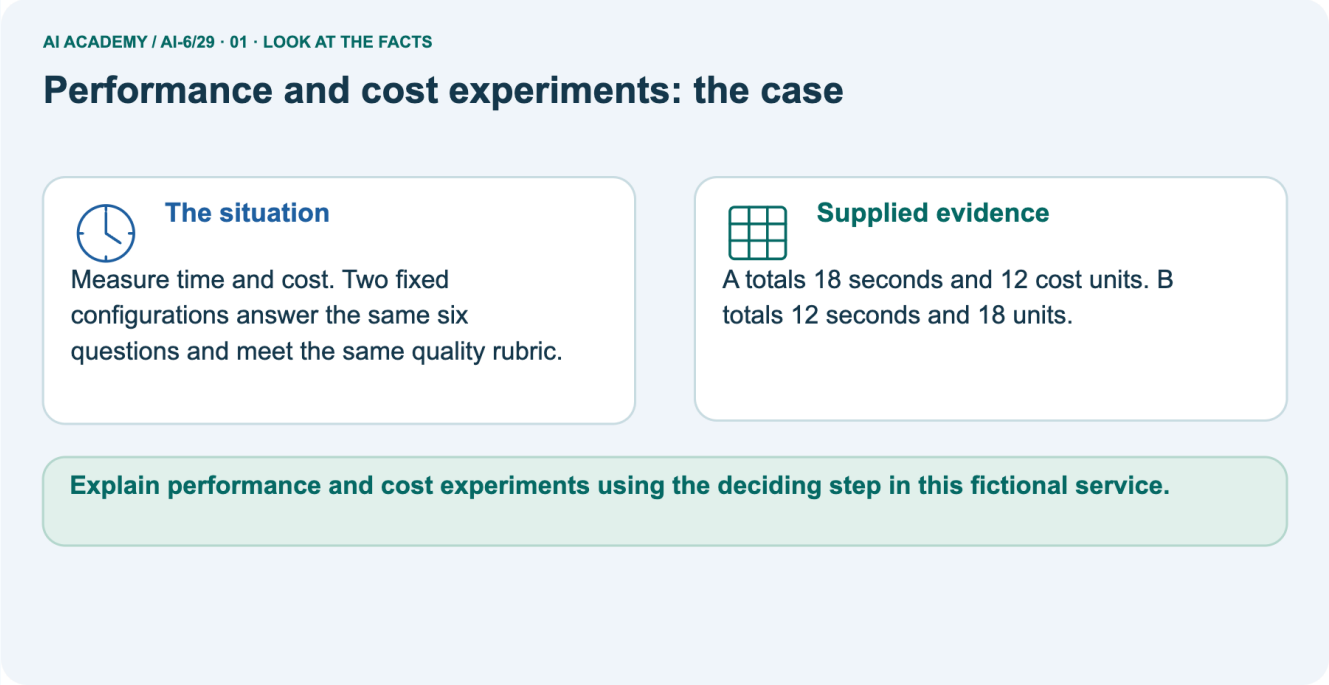
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Performance and cost experiments: the case



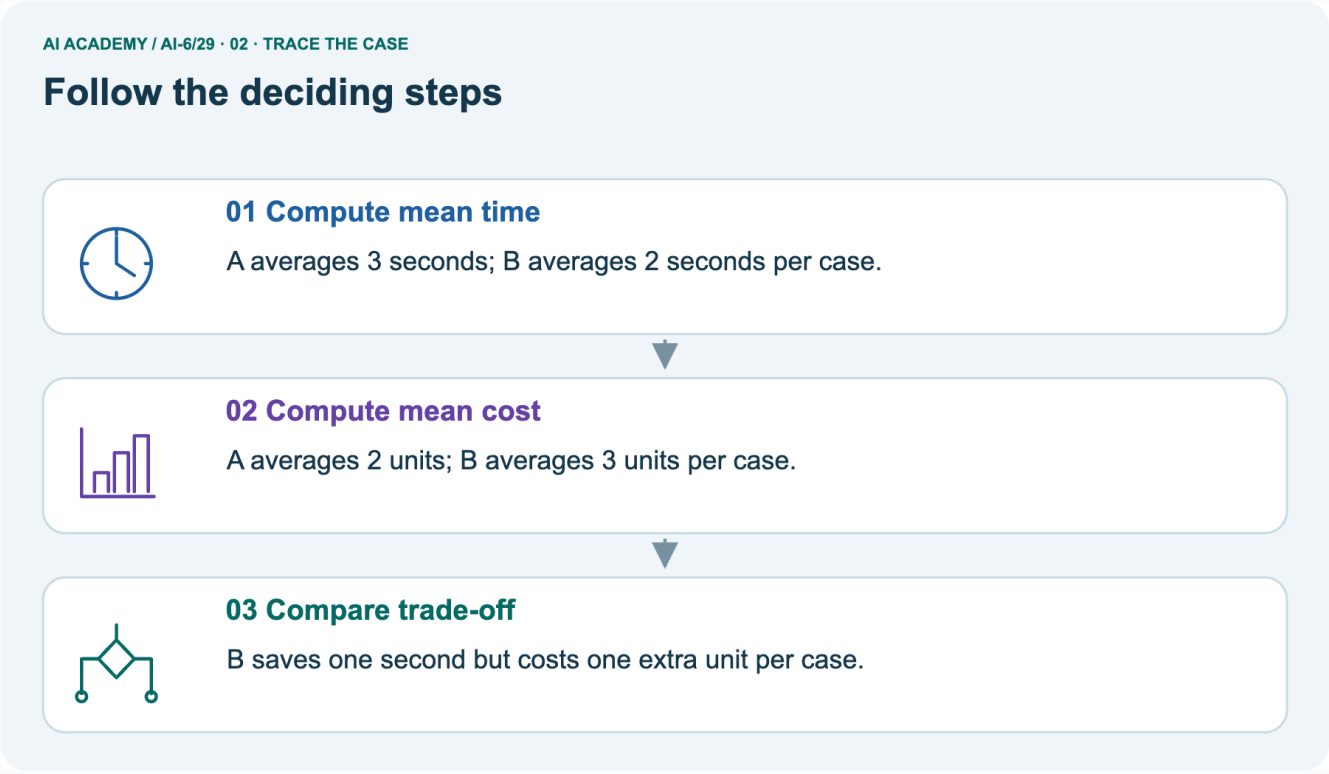
Follow the labelled picture with the matching explanation. Open the vector diagram

Measure time and cost. Two fixed configurations answer the same six questions and meet the same quality rubric. Supplied facts: A totals 18 seconds and 12 cost units. B totals 12 seconds and 18 units.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



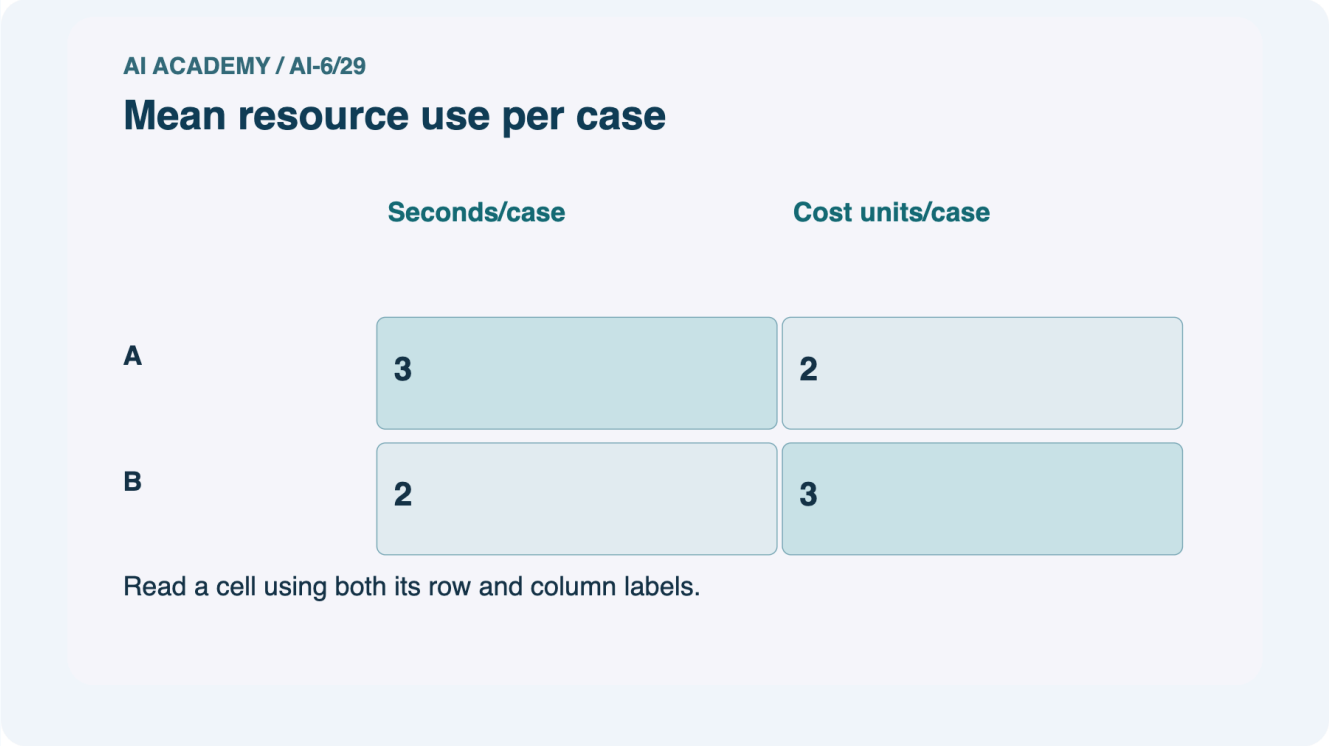
Follow the labelled picture with the matching explanation. Open the vector diagram

Compute mean time: A averages 3 seconds; B averages 2 seconds per case. → Compute mean cost: A averages 2 units; B averages 3 units per case. → Compare trade-off: B saves one second but costs one extra unit per case.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Mean resource use per case



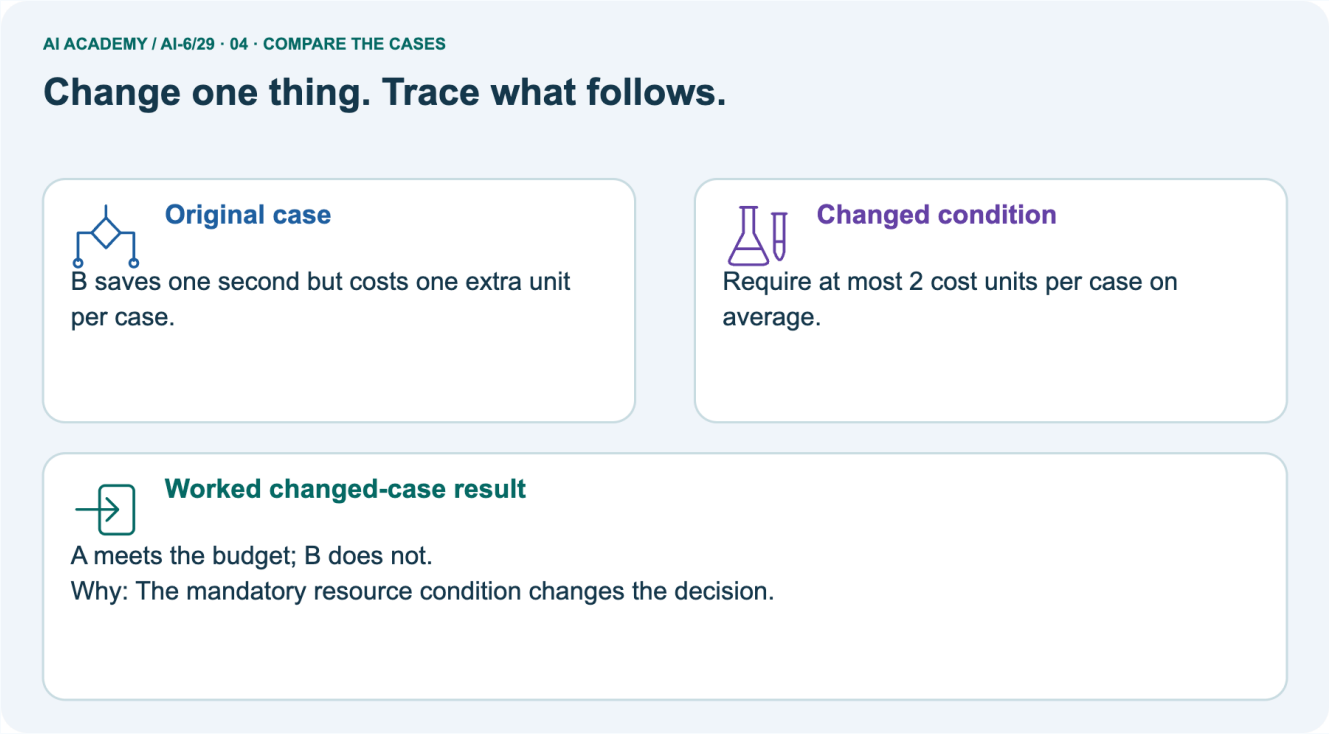
Follow the labelled picture with the matching explanation. Open the vector diagram

Mean resource use per case. A: Seconds/case = 3, Cost units/case = 2. B: Seconds/case = 2, Cost units/case = 3.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



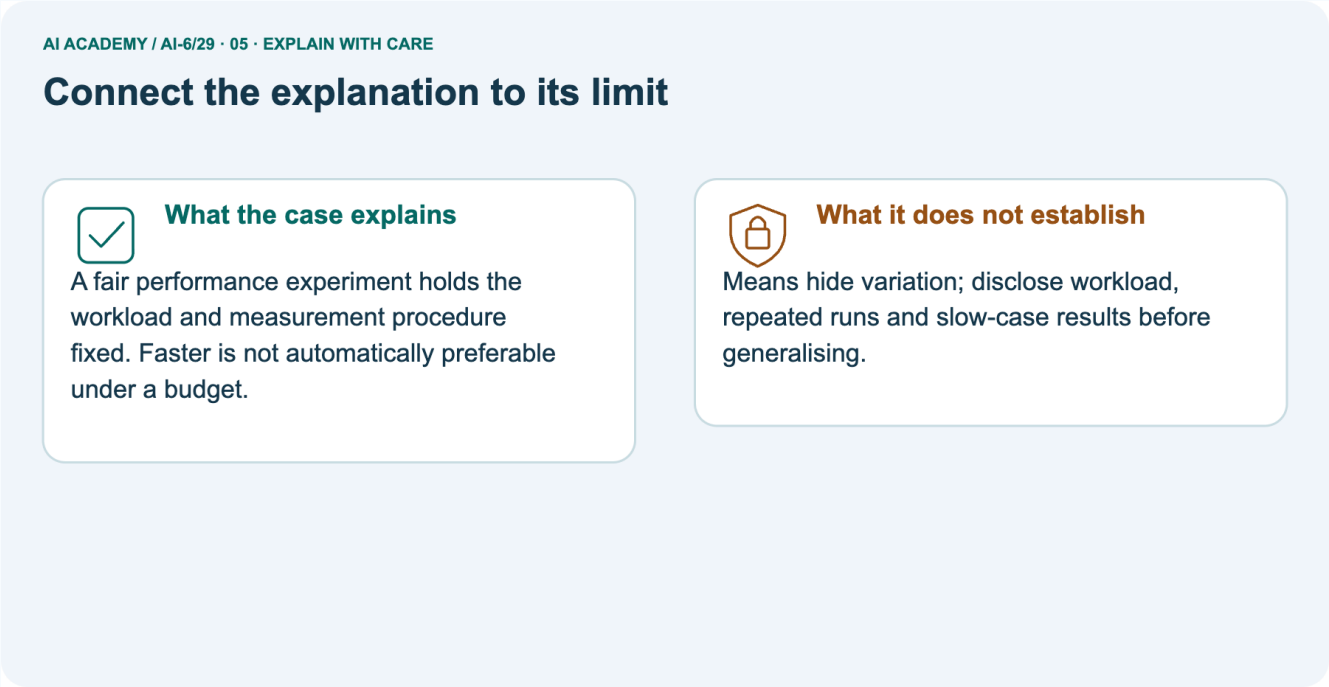
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: B saves one second but costs one extra unit per case. Changed condition: Require at most 2 cost units per case on average. Expected result: A meets the budget; B does not. Reason: The mandatory resource condition changes the decision.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

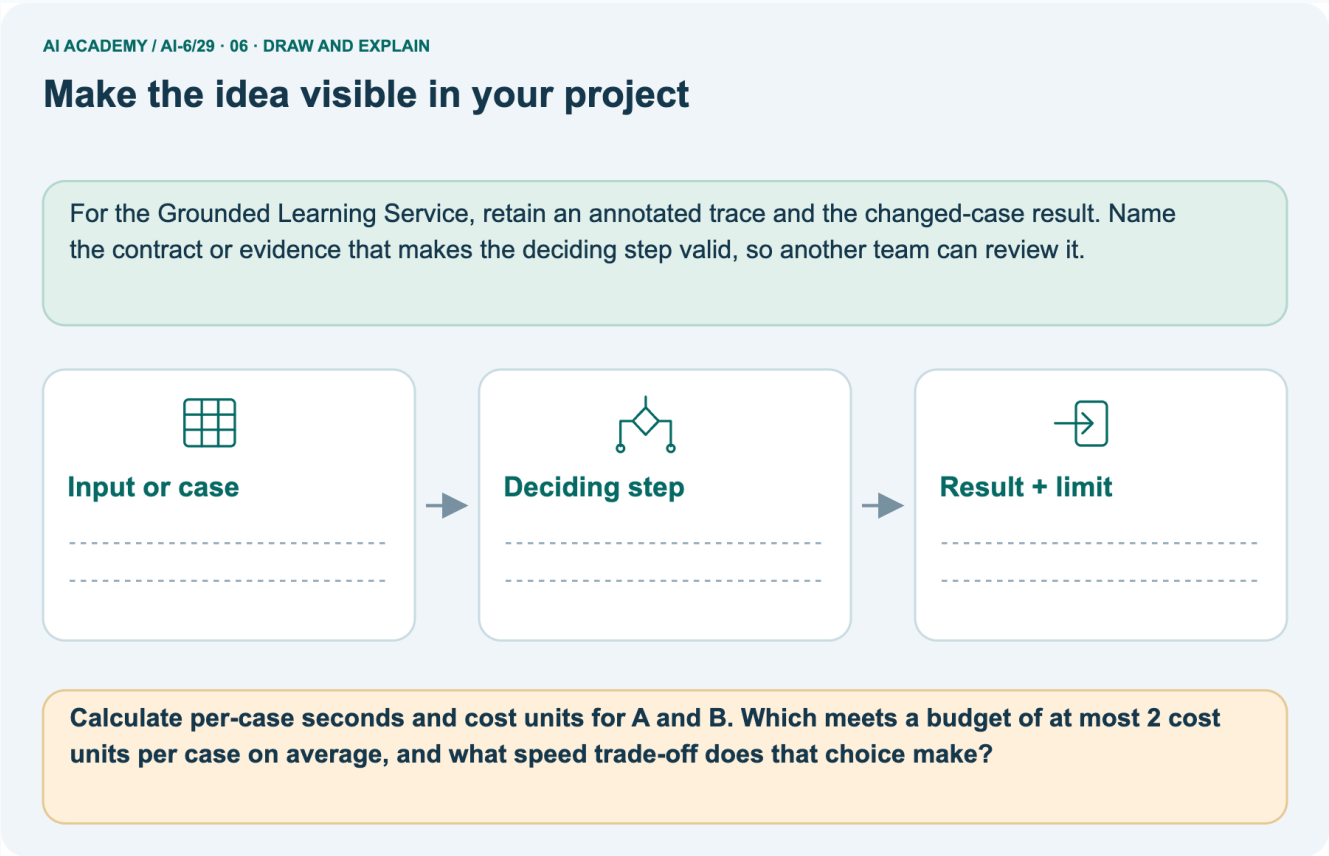
A fair performance experiment holds the workload and measurement procedure fixed. Faster is not automatically preferable under a budget.

Limit: Means hide variation; disclose workload, repeated runs and slow-case results before generalising.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it. Draw your **input**, deciding step, result and limit. Supported question: Calculate per-case seconds and cost units for A and B. Which meets a budget of at most 2 cost units per case on average, and what speed trade-off does that choice make?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Performance and cost experiments: the case

02. Follow the deciding steps

03. Mean resource use per case

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.